

DBMS — Lecture 2

Features of ER Model

Learning Objectives

Understand major features of the ER model.

Understand cardinality, participation, keys, and weak entities.

Read a simple ER diagram.

Major Features

Entities and Entity Sets: Represent real-world objects and collections of similar objects.

Attributes: Describe properties of entities or relationships.

Relationships: Represent associations between entity sets.

Keys: Uniquely identify entities.

Cardinality: Specifies how many entities can participate in a relationship.

Participation: Indicates whether participation is mandatory or optional.

Cardinality Types

1:1: One entity is associated with at most one entity on the other side.

1:N: One entity can be associated with many entities.

M:N: Many entities can be associated with many entities.

Participation

Total participation: Every entity must participate in the relationship.

Partial participation: Participation is optional for some entities.

Advanced Features

Weak Entity: Cannot be uniquely identified by its own attributes alone and depends on an owner entity.

Specialization/Generalization: Represents superclass-subclass relationships.

Relationship attributes: Attributes that belong to a relationship, such as Enrollment_Date.

Example

STUDENT and COURSE have an M:N relationship called ENROLLS. ENROLLS may have attributes such as Semester and Enrollment_Date.

Key Takeaway

ER features help capture structure, constraints, and business rules before database implementation.